Exp 11a – Synthesis of Alum

INTRODUCTION

In this experiment, we will synthesize potassium aluminum sulfate dodecahydrate ($KAl(SO_4)_2 \cdot 12H_2O$) from solid aluminum. We will use stoichiometry to predict the mass of alum we can make. From the experimental results, it is possible to calculate the % yield of this reaction.

BACKGROUND

The reaction to form Alum from aluminum takes place in the following steps.

$$2Al(s) + 2KOH(aq) + 6H_2O(l) \rightarrow 2KAl(OH)_4(aq) + 3H_2(g) \quad (10)$$

$$2KAl(OH)_4(aq) + H_2SO_4(aq) \rightarrow 2Al(OH)_3(s) + K_2SO_4(aq) + 2H_2O(l) \quad (11)$$

$$2Al(OH)_3(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 6H_2O(l) \quad (12)$$

$$Al_2(SO_4)_3(aq) + K_2SO_4(aq) + 24H_2O(l) \rightarrow 2KAl(SO_4)_2 \cdot 12H_2O(s) \quad (13)$$

These equations can be combined to give the following total chemical equation for the process.

$$2Al(s) + 2KOH(aq) + 4H_2SO_4(aq) + 22H_2O(l) \rightarrow 2KAl(SO_4)_2 \cdot 12H_2O(s) + 3H_2(g) \quad (14)$$

<u>Some interesting facts about Alum</u>

Potassium aluminum sulfate dodecahydrate, or "Alum" for short, has been collected and processed for centuries. It is used in industry as a water clarifier (in water purification), in the manufacture and application of dyes, in the tanning of leather, and in marble and porcelain cements. It is also the ingredient in pickles that gives them their crispness. Medicinally it is used as an astringent (a substance that draws together or constricts tissues) and a styptic (to control bleeding). Alum is used extensively in the paper industry as a sizing agent (a material that fills the spaces between the paper fibers to control its porosity and to bind other additives).

ADDITIONAL READING

- Stoichiometry and Yield Calculations

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

- 50-mL beaker
- 150, 200, or 250-mL beaker
- 600-mL beaker
- 10-mL graduated cylinder
- 100-mL graduated cylinder
- Transfer pipets
- Glass-stirring rod
- Spatula
- Medium watch glass
- Büchner funnel
- Side-arm flask
- Filter paper
- Hot Plate

CLASSROOM SUPPLY AND REACTANT LIST

- KOH (1.5 M)
- H_2SO_4 (6.0 M)
- Alcohol solution (50 %)
- Aluminum foil or powder
- Ice
- Vacuum hose
- Bench stand
- Clamp

PROCEDURE

** Warning: both potassium hydroxide and sulfuric acid are very corrosive and are very damaging to skin, eyes, clothes, etc. Be very careful when working with or around them! **

1. ☐ Record the mass of a clean *dry* small beaker (150, 200 or 250-mL).
2. ☐ Obtain approximately 0.4 g aluminum foil and add it to the beaker by tearing or cutting it into small pieces (about the size of a pencil eraser)
3. ☐ Record the mass of the beaker with the aluminum foil.
4. ☐ Label your beaker with your name with a permanent marker (Sharpie ®).
5. ☐ Using a graduated cylinder, add 20 mL of 1.5 M KOH to the beaker, while in the fume hood **. The aluminum may take about 20 minutes to completely react with the KOH. (It has completely reacted when it is a black solution and there are no pieces of aluminum foil left.) During this time, calculate your theoretical yield of alum.

   ** The hydrogen gas emitted can generate a fine mist of corrosive potassium hydroxide. **

   * If the reaction slows, and does not seem to be proceeding, you may heat up the beaker briefly on a hot plate but be careful to ensure that the solution does not boil.
6. ☐ When the aluminum has completely reacted, obtain 7.5 mL of 6.0 M H_2SO_4 from a bottle-pump dispenser into a 50-mL beaker (one complete pump).
7. ☐ Using a transfer pipet, add the acid dropwise to the reaction mixture, while stirring with a glass-stirring rod.

   * After the addition of the acid, the reaction mixture is usually greyish white.

8. ☐ Cover the beaker with a watch glass and warm the solution on a hot plate until the solid has completely dissolved. (The solution will be a transparent tan color with no visible solid chunks)
   ** Do not boil the solution! Hot concentrated sulfuric acid is very corrosive. **
9. ☐ When the solid has completely dissolved, set the beaker aside for a few minutes to cool.
10. ☐ While the beaker is cooling, either locate the ice bath set up by your instructor or make one yourself by putting ice and water into a beaker larger than the one you are using for the reaction mixture.
11. ☐ Once the beaker has cooled a little, place it in the ice bath and allow it to cool for 15 minutes.
12. ☐ Swirl the beaker when you notice the onset of crystal formation. If alum crystals are not starting to form within 5 minutes, it may be necessary to scratch the inside walls of the beaker with a glass stirring rod. This provides sites for crystallization to begin.
13. ☐ While the solution is cooling, pour 30 mL of 50 % alcohol (ethanol) into a small beaker and place it in the ice bath. The alcohol will be used to rinse your crystals out of the beaker during the filtration step. (If your instructor has the alcohol solution in ice, you can wait until you are ready to filter to obtain it)
14. ☐ Set up the vacuum filtration apparatus as shown in Figure 1, if it is not already set up in the classroom.

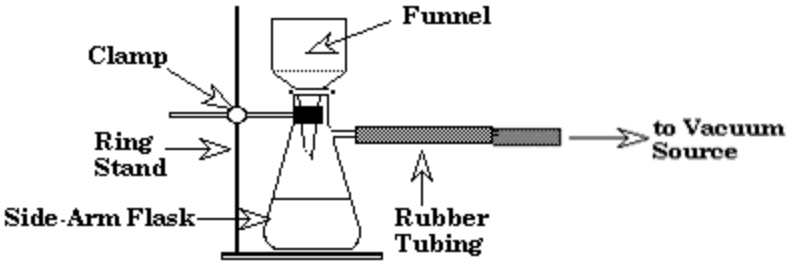


Figure 1: Vacuum Filtration Apparatus

15. ☐ Take your beaker with alum crystals and 50 % alcohol solution to the vacuum apparatus, then:
    a. Turn the vacuum on so that air is being drawn through the *filter paper*. Leave the vacuum on through the entire process below.
       * Make sure that all the holes in the funnel are covered. Dampen the filter paper so that it stays in place.
    b. Swirl the beaker so that all the crystals are dislodged and dump the entire contents of the beaker quickly into the center of the filter paper within the Buchner funnel.
    c. Pour about 10 mL of the cooled 50 % alcohol solution into the beaker that contains the alum, then swirl the alum beaker and pour the mixture into the funnel to transfer any remaining crystals while washing the crystals already in the funnel. (Repeat until you have used all the alcohol solution.)
    d. Leave the vacuum on for a few minutes until the alum crystals look somewhat dry, then turn off the vacuum. You can gently poke at it with a stir rod to see how dry it is.

16. ☐ Obtain the mass of a clean dry watch glass, then transfer the alum crystals to the watch glass and allow them to continue to air dry.
17. ☐ Obtain a final mass for the watch glass and alum.
18. ☐ Record your experimental (actual) yield and calculate your percent yield for the synthesis of alum.
19. ☐ Discuss your percent yield. (Does this number seem reasonable? Is this what you would expect?)

CLEANUP AND WASTE DISPOSAL

1. ☐ Dispose of the alum crystals in the solids-waste container.
2. ☐ Dispose of the waste from the filtration step (in the side-arm flask) in the inorganic aqueous waste container.
3. ☐ Dispose of the filter paper and plastic transfer pipet in the trash can.
4. ☐ Use acetone to wipe off permanent marker on equipment.
5. ☐ Wash all glassware with soapy water. Rinse with DI water. There is no need to thoroughly dry the equipment prior to returning it to the shelves.

CALCULATIONS

1. ☐ Determine the mass of aluminum added to the reaction beaker.
2. ☐ Based on the mass of aluminum (Al), calculate the theoretical yield of alum ($KAl(SO_4)_2 \cdot 12H_2O$) that could be made:

$$2Al(s) + 2KOH(aq) + 4H_2SO_4(aq) + 22H_2O(l) \rightarrow 2KAl(SO_4)_2 \cdot 12H_2O(s) + 3H_2(g)$$

3. ☐ Calculate the mass of alum you actually obtained in the experiment.
4. ☐ Calculate the percent yield:

$$Percent\ Yield = \frac{Actual\ Yield}{Theoretical\ Yield} \times 100$$

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

REFERENCE VALUES FROM PRE-LAB QUIZ

2. Molar Mass of aluminum _____

3. Molar Mass of Alum _____

4. Reaction ratio _____mol Al : _____ mol $KAl(SO_4)_2 \cdot 12H_2O$

DATA

1. Mass of the reaction beaker _____

5. Mass of the beaker + aluminum _____

6. Mass of watch glass _____

7. Mass of watch glass + alum _____

RESULTS

Show your work for each of the following calculations.

8. Mass of aluminum _____

9. Theoretical yield of Alum _____

10. Mass of Alum _____ (actual yield)

11. Percent yield _____

12. Discuss your percent yield. (Is this a reasonable value? Why or why not?)